# MILITARY SPECIFICATION

## SYRINGE KIT, DENTAL IMPRESSION AND IRRIGATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a disposable multipurpose syringe suitable for use with elastomeric dental impression material and irrigating solutions.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | Boxes, Shipping, Fiberboard. |
| PPP-B-676 | Boxes, Set-up. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila., Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-147 | Palletized Unit Loads. |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging of. |

DRAWING

Defense Personnel Support Center

| | |
|---|---|
| 24501 | Syringe, Dental, Impression and Irrigating. |

(Copies of specifications and standards, required by contractors in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

DEPARTMENT OF HEALTH AND HUMAN SERVICES

FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 Material. All material shall comply with the requirements of the Federal Food, Drug and Cosmetic Act. The syringe, except for extender, shall be physically and chemically compatible and inert with standard irrigating solutions, or elastomeric impression materials. The syringe barrel and nozzle shall be fabricated of medical grade polypropylene plastic. The nozzle material shall facilitate cutting so that both length and oriface diameter are thereby adjustable. The plastic sheets shall be fabricated of clear, transparent polystyrene. The plunger rod shall be fabricated of opaque, medical grade, polypropylene plastic. The plunger diaphragm shall be fabricated of black pharmaceutical grade rubber compound. The finger grip extenders shall be fabricated of polypropylene plastic. The lubricant shall be a silicone base, inert and non-toxic, medical fluid.

3.2 Design and construction. The syringe shall be a disposable barrel and plunger type. The syringe shall conform in style, dimensions, and construction to Defense Personnel Support Center (DPSC) Drawing 24501. The syringe shall consist of a barrel and a curved dispenser nozzle, plunger rod and finger grip extenders. The syringe shall be suitable for use with elastomeric impression materials and irrigating solutions.

3.2.1 Barrel. The barrel shall be constructed with the curved tip molded to the body of the barrel. Finger grips shall be flattened on two sides to prevent rolling. Base of curved tip shall be convex to interface with convex shape of rubber diaphragm.

3.2.2 Plunger rod. Plunger rod shall be a one-piece molding, designed to withstand required pressures when used as intended. Mesial end of rod shall be provided with a thumb rest of prescribed diameter to permit firm, comfortable manipulation. Distal portion of rod shall be designed to receive and hold the plunger diaphragm in precise perpendicular relationship to the walls of the barrel. Distal tip of plunger rod shall be provided with a projection which shall insure complete seating of the diaphragm into the funnel-shaped portion of the barrel.

3.2.3 Plunger diaphragm. Plunger diaphragm shall be a flexible, double-finned, conical tipped, one-piece molding, designed to fit onto the plunger rod and to seat firmly into the funnel shaped portion of the barrel when used as intended. Diaphragm shall be lightly coated with a lubricant and shall be in position on the plunger rod when furnished.

3.2.4 Finger grip extenders. Reusable finger grip extenders shall be provided to permit application of increased pressure where required. Finger grip extender shall consist of a one-piece molding, so designed that it may be easily placed over the barrel, against the finger grip of the barrel, and supported upon the barrel by four projections. Finger grip extenders shall be readily removable.



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3.2.5 Plastic sheets. Plastic sheets, suitably packaged for protection against wrinkling, shall be provided for use in loading the syringe. Each sheet shall be 0.005 ± 0.0015 inch in thickness and shall measure 3-1/8 ± 1/8 by 2 ± 3/16 inches.

3.3 Finish. The syringe barrel and nozzle shall have a smooth surface finish to prevent any irritations to the mouth tissues.

3.4 Marking. Each syringe shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.5 Quantity. Each unit package shall consist of fifty disposable 12 ml syringes, fifty plastic sheets and two finger grip extenders.

3.6 Instructions. Each unit package shall be furnished with a sheet of accurate and adequate instructions describing the method used in depositing the impression material on plastic sheet, rolling the sheet around material to form a cylinder smaller than barrel diameter and inserting the cylinder into the barrel of syringe. As an alternate, the instructions may be printed on the unit box.

3.7 Workmanship. The syringe shall be clean and free from defects which detract from its appearances or impair its serviceability.

3.8 Autoclavability. The syringes shall show no adverse effects such as warping, deformation or binding when tested as specified in 4.4.2.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.



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4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105 and, unless specified, shall be on the unit of product being one syringe.

4.2.2 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection and acceptable quality level (AQL) shall be as indicated in table I.

Table I. Sampling for Examination

| Type of Defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major A | II | 1.0 |
| Major B | S-2 | 1.5 |
| Minor A | II | 2.5 |
| Minor B | S-2 | 4.0 |

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-3; acceptance number shall be zero for all sample sizes.

4.3 Examination. The syringe shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and II. Inspection, however, shall not be restricted to defects listed in table II. Any syringe in the sample containing one or more defects shall be rejected and, if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.



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TABLE II. Classification of Defects

| Categories | Defects |
|---|---|
| Critical: | None defined. |
| Major A: | |
| 101 | Nozzle bore not free from obstructions. |
| 102 | Nozzle surface not free from sharp edges or parting lines. |
| 103 | Plunger rod does not move freely in barrel. |
| 104 | Barrel tip orifice blocked. |
| 105 | Plunger diaphram detaches from plunger rod when rod is pushed back and forth in barrel. |
| 106 | Barrel shows evidence of holes, cuts, gouges, cracks. |
| 107 | Nozzle not in specified alignment with barrel. |
| 108 | Plunger diaphragm not made of pliable rubber. |
| 109 | Plunger rod frozen in barrel. |
| Major B: | |
| 151 | Syringe dimensions affecting fit of plunger rod in barrel not as specified. |
| 152 | Marking on immediate container missing, not legible, not correct, or not complete. |
| 153 | Nozzle not securely molded to barrel. |
| 154 | Instructions omitted. |
| Minor A: | |
| 201 | Nozzle not in specified alignment with barrel. |
| 202 | Syringe finish not as specified. |
| Minor B: | |
| 251 | Unit package contains incorrect count of syringes. |
| 252 | Dimensions (other than 151) not as specified. |

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operational test. The operational test shall be performed by moving the plunger rod back and forth, five times, in the barrel and checking for freedom of movement, binding, or freezing. The plunger diaphragm shall be tested for a tight fit on the plunger rod to insure its beirg retained during the backward and forward movement of the rod in the barrel.



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4.4.2 Autoclaving. The syringes shall be autoclaved at 250°F ± 5° and 15 ± 1 pounds per square inch for 30 minutes. After the syringes are removed from the autoclave, they shall be examined for evidence of adverse effects such as warping, deformation or binding.

4.5 Inspection of packaging. The sampling and inspection of the packaging and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specifications and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each syringe kit, consisting of fifty syringes enclosed in a clear plastic bag, fifty plastic sheets neatly inserted in an envelope, and two finger grip extenders enclosed in an envelope or bag, shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the bags, envelopes and box shall be effected by the use of tape, staples or glueing.

5.1.2 Level C. Kit shall be packaged in accordance with the requirements of MIL-STD-794.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Ten (10) kits shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of the appendix to MIL-L-10547. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the applicable box specifications.

5.2.2 Level B. Ten (10) kits shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Ten (10) kits shall be packed in accordance with MIL-STD-794.



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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Kits packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.4.4.2 Level C. Kits packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Unit package. Unit package shall be marked in accordance with MIL-STD-129.

5.3.2 Exterior container. Exterior container shall be marked as specified in MIL-STD-129.

5.3.3 Unitized load. Unitized load shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6520-01-173-2081 | SYRINGE KIT, Dental Impression and Irrigating, Barrel and Plunger Syringe, Disposable, 12 ml |



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MILITARY INTERSTS:

Custodians
Army - MD
Navy - MS
Air Force - 03

Review activities

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITY:
VA-OSS

53526

Preparing Activity:

DoD-MD

Agent:

DLA-DM

Project No. 6520-1827

Location: NCR



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-42009 | Syringe Kit, Dental Impression and Irrigating |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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